A Hierarchical Gene–Isoform Model for Sparse Single-Cell Data

Purpose

The existing genome-wide factorization models normalized equivalence-class (EC) counts directly. It recovers cell-type information, but a matched evaluation shows a substantial gap from a standard gene-expression analysis. Increasing the factor rank does not close this gap. Adding an auxiliary gene-reconstruction loss helps, indicating that the EC objective spends too much capacity on transcript ambiguity and too little on marker-gene abundance.

This note proposes a hierarchical count model that separates gene abundance from conditional isoform allocation. The model is designed for cells that do not contain enough molecules for independent transcript-level EM. Statistical strength instead comes from pooled transcript baselines, low-dimensional cell states, and transcript loadings shared across all cells.

Notation

Let $m = 1, \ldots, M$ index cells, $g = 1, \ldots, G$ genes, and $t = 1, \ldots, T$ transcripts. Let $g(t)$ be the gene containing transcript t, and let $\mathcal{T}_g = \{t : g(t) = g\}$. Primer or library type is indexed by p. For primer p, let

$$A_p \in \mathbb{R}_+^{K_p \times T}$$

be the fixed EC-by-transcript observation matrix. It includes alignment compatibility, Salmon conditional weights and positional-bias correction, and the primer-specific transcript sampling exposure. In particular, the primary paired model uses no additional transcript-length exposure for oligo(dT) and relative effective-length exposure for random-hexamer priming.

Let n_{kmp} be the observed UMI count for EC k, cell m, and primer p, and let $N_{mp} = \sum_k n_{kmp}$.

Hierarchical Abundance

Transcript abundance is decomposed as

$$\theta_{tm} = a_{g(t)m}\pi_{tm}, \qquad \sum_{t \in \mathcal{T}_g} \pi_{tm} = 1.$$

Here $a_{gm} > 0$ is the abundance of gene g in cell m, while π_{tm} is the conditional fraction of gene $g(t)$'s molecules assigned to transcript t.

This within-gene normalization is not a global simplex constraint on Θ. It is needed to identify gene abundance separately from isoform scale. The gene abundances may remain unnormalized because the observation likelihood below normalizes predicted EC mass within each cell and primer.

Gene abundance factors

Let $\boldsymbol{z}_m \in \mathbb{R}^{r_g}$ be the gene-expression state of cell m. Model gene abundance as

$$\log a_{gm} = \alpha_g + \boldsymbol{u}_g' \boldsymbol{z}_m,$$

where α_g is a pooled gene intercept and $\boldsymbol{u}_g$ is a gene loading. The cell state $\boldsymbol{z}_m$ is the natural representation for cell annotation, visualization, and comparison with standard gene-expression analyses.

An exponential link is convenient for a count model. A softplus link can be used if bounded gradients are more stable:

$$a_{gm} = \mathrm{softplus}(\alpha_g + \boldsymbol{u}_g' \boldsymbol{z}_m).$$

Conditional isoform factors

Let β_t be the pooled logit for transcript t. The initial model uses the gene-expression state to explain isoform usage:

$$\eta_{tm} = \beta_t + \boldsymbol{w}_t' \boldsymbol{z}_m, \qquad \pi_{tm} = \frac{\exp(\eta_{tm})}{\sum_{j \in \mathcal{T}_{g(t)}} \exp(\eta_{jm})}.$$

The transcript loading $\boldsymbol{w}_t$ describes isoform changes associated with the same cell state that explains gene expression. This is preferable to estimating a second free cell representation immediately: low-count cells can then borrow isoform information from cells with similar gene-expression profiles.

If held-out molecules support additional isoform structure, introduce a smaller isoform-specific state $\boldsymbol{h}_m \in \mathbb{R}^{r_i}$:

$$\eta_{tm} = \beta_t + \boldsymbol{w}_t' \boldsymbol{z}_m + \boldsymbol{r}_t' \boldsymbol{h}_m.$$

The $\boldsymbol{r}_t' \boldsymbol{h}_m$ term captures isoform programs that are not predictable from gene expression. The nested model $r_i = 0$ should be the default, and $r_i > 0$ should be accepted only by label-blind molecule-count cross-validation.

Paired-Primer Count Likelihood

For a cell and primer, define the unnormalized expected EC mass

$$\boldsymbol{q}_{mp} = A_p \boldsymbol{\theta}_m.$$

Conditioning on the observed UMI total gives

$$\boldsymbol{n}_{mp} \mid N_{mp} \sim \mathrm{Multinomial}\left(N_{mp}, \frac{\boldsymbol{q}_{mp}}{\mathbf{1}' \boldsymbol{q}_{mp}}\right).$$

Ignoring constants, the negative log likelihood is

$$\mathcal{L} = -\sum_{m,p} \sum_{k:n_{kmp}>0} n_{kmp} \log q_{kmp} + \sum_{m,p} N_{mp} \log(\mathbf{1}' \boldsymbol{q}_{mp}).$$

The two primer halves share $\boldsymbol{z}_m$, $\boldsymbol{h}_m$, a_{gm}, and π_{tm}, but use separate A_p. Conditioning on N_{mp} removes a primer-by-cell exposure parameter and preserves the current convention of normalizing each half independently.

An independent Poisson model,

$$n_{kmp} \sim \mathrm{Poisson}(s_{mp} q_{kmp}),$$

is an alternative when total UMI information is intended to contribute to fitting. It requires estimating or profiling the exposure s_{mp}. The conditional multinomial model is the simpler first implementation and directly matches the sampling unit used for molecule-count folds.

Why Cell-Level EM Is Not Required

Independent EM would estimate a high-dimensional transcript vector in every cell. That is underdetermined for cells with hundreds or a few thousand UMIs. The hierarchical model instead estimates only low-dimensional cell coordinates:

$$\boldsymbol{z}_m \in \mathbb{R}^{r_g}, \qquad \boldsymbol{h}_m \in \mathbb{R}^{r_i}.$$

All gene and transcript intercepts and loadings are shared across cells. Cells without isoform-informative molecules remain close to the pooled baseline β_t, while informative cells update the shared relationship between cell state and isoform usage.

EM may still be useful on pooled data to initialize β_t. A global or pseudobulk EM estimate has enough molecules to establish baseline isoform usage, but it is not treated as a cell-level estimator. Joint minibatch gradient optimization of the count likelihood then estimates all cell states and shared loadings.

Initialization and Fitting

A staged fit reduces non-convexity and provides useful diagnostics:

1. Aggregate ECs that map entirely within one gene and fit $\alpha_g, \boldsymbol{u}_g, \boldsymbol{z}_m$ with a gene-level count factor model. Cross-gene ambiguous ECs are excluded only from this initializer.
2. Estimate pooled β_t from all cells or sufficiently large pseudobulks. Pooled EM is acceptable at this stage.
3. With $\boldsymbol{z}_m$ initialized, fit $\boldsymbol{w}_t$ under the complete EC likelihood while holding $r_i = 0$.
4. Jointly fine-tune gene abundance, conditional isoform usage, and both primer observation models.
5. Fit candidate $r_i > 0$ models by adding small isoform-specific factors $\boldsymbol{h}_m, \boldsymbol{r}_t$. Warm-start from smaller to larger r_i.

The fit should use Adam or another stable first-order optimizer initially, followed by deterministic full-data or large-batch polishing. Gene and isoform cell factors should use separate learning rates because isoform gradients are supported by fewer molecules. Gradient clipping and log-sum-exp evaluation are required for numerical stability.

Weak Gaussian priors or equivalent quadratic penalties on $\boldsymbol{h}_m$ and $\boldsymbol{r}_t$ are appropriate even though the previous direct factorization did not use an L_2 penalty. Here they control an explicitly optional residual state that would otherwise fit sparse-cell noise. Their strength, or the decision to omit $\boldsymbol{h}_m$, must be selected from held-out molecules rather than labels.

Scalable Likelihood Evaluation

The likelihood need not construct a dense cell-by-EC matrix. For a cell minibatch, only ECs with nonzero counts contribute to $\sum_k n_{kmp} \log q_{kmp}$. Each such value uses the sparse transcript list stored in the corresponding row of A_p.

The normalizing term is also inexpensive once transcript abundance is available. Define the transcript column-sum vector

$$\boldsymbol{d}_p = A_p' \mathbf{1}.$$

Then

$$\mathbf{1}' A_p \boldsymbol{\theta}_m = \boldsymbol{d}_p' \boldsymbol{\theta}_m,$$

without constructing all predicted EC values.

For a minibatch of B cells, the main dense workspace is the $T \times B$ transcript abundance block. Batches of a few hundred cells keep this bounded even for a genome-wide transcriptome. Observed EC likelihood terms can be evaluated from a COO event table containing cell, primer, EC, and count, joined to the sparse EC-to-transcript edges. Segmented reductions compute each observed q_{kmp}.

Explicit cell-factor storage remains small. For example, one million cells with a 32-dimensional float32 gene state require approximately 128 MB; an additional eight-dimensional isoform state requires approximately 32 MB. An amortized encoder from gene counts to $\boldsymbol{z}_m$ is a later option, but is not required for the initial million-cell implementation.

Identifiability

The following conventions remove avoidable invariances:

- Center the gene intercepts α_g, because a common shift of all gene log abundances is removed by the conditional multinomial likelihood.
- Center β_t within each gene because adding the same value to every transcript logit leaves $\boldsymbol{\pi}_{gm}$ unchanged.
- Center each column of $\boldsymbol{w}_t$ and $\boldsymbol{r}_t$ within each gene for the same reason.
- Center and standardize cell-factor columns after each epoch, absorbing location and scale changes into intercepts and loadings.
- Use factor balancing or an orthogonality convention for reporting. Latent rotations do not affect fitted abundance, but a stable convention improves warm starts and interpretation.

These constraints are parameter-identification conventions, not biological regularization and not simplex constraints on the genome-wide transcript matrix.

Label-Blind Model Selection

Molecule counts should be partitioned before any normalization, as in the existing count-fold procedure. Candidate models are compared using held-out multinomial log likelihood on the same response scale.

The recommended nested path is:

$$r_i = 0, 2, 4, 8, 16,$$

with small-to-large warm starts. The gene rank r_g can be selected first, then held fixed while selecting r_i. A one-standard-error rule should prefer $r_i = 0$ or the smallest isoform rank statistically indistinguishable from the minimum held-out loss.

Held-out diagnostics should also decompose performance into:

- gene-unambiguous molecule likelihood;
- within-gene EC likelihood;
- cross-gene ambiguous EC likelihood;
- poly(dT) and random-hexamer likelihoods.

This distinguishes genuine improvement in isoform prediction from improvement that comes only through gene abundance. Cell-type labels, ARI, NMI, and silhouette remain external assessments and are never hyperparameter inputs.

Downstream Differential Isoform Analysis

The fitted π_{tm} is the conditional isoform usage quantity needed downstream. Differential effects can be introduced directly in the isoform logits:

$$\eta_{tm} = \beta_t + \boldsymbol{w}_t' \boldsymbol{z}_m + \boldsymbol{r}_t' \boldsymbol{h}_m + \boldsymbol{x}_m' \boldsymbol{\gamma}_t,$$

where $\boldsymbol{x}_m$ contains diagnosis, cell type, donor covariates, and interactions. Coefficients $\boldsymbol{\gamma}_t$ are centered within each gene. Inference should respect donor replication, either through donor-level random effects or by fitting and testing on donor-by-cell-type pseudobulks of the model-implied sufficient statistics.

This preserves the intended division of labor: gene factors provide a robust cell representation, shared isoform loadings borrow information across cells, and downstream tests operate on transcript-resolved conditional usage rather than requiring independent cell-level EM estimates.

Recommended First Implementation

The first implementation should deliberately omit the free isoform cell state:

$$\log a_{gm} = \alpha_g + \boldsymbol{u}_g' \boldsymbol{z}_m, \qquad \pi_{tm} = \mathrm{softmax}_{t \in \mathcal{T}_{g(t)}} (\beta_t + \boldsymbol{w}_t' \boldsymbol{z}_m).$$

It should use the paired-primer multinomial EC likelihood, pooled-EM initialization of β_t, explicit cell factors, and GPU minibatches. Only after this model is converged and evaluated should an independent $\boldsymbol{h}_m$ be added. This ordering tests the central hypothesis—that robust gene state can support transcript-resolution inference—without giving every sparse cell enough freedom to reproduce its EC noise.